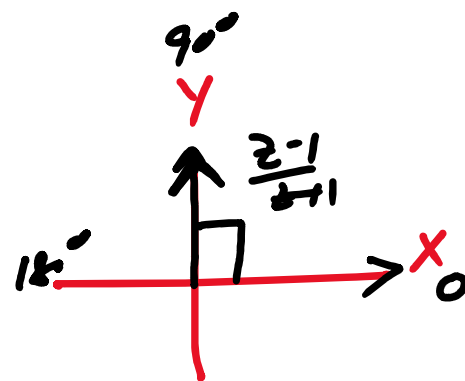


Mathematics Class 11 Federal
Board, 2026 Model Paper Sec B
& C (Short + Long) Live Session
Day # 03, 9 Days Left

[Short + Long] → Guels paper ↓ ≡
 Ex Ex ←→ (60%)
 ←→

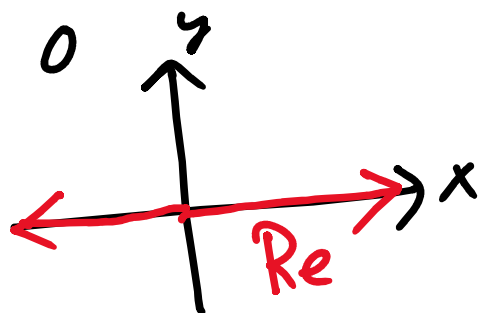
If $z = x + iy$ and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$, then
 show that $x^2 + y^2 = 1$

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$$



Pure imaginary $z = 0 + iy$
 ↓
 Real part.

$$\arg\left(\frac{z-1}{z+1}\right) = 0$$



$$\bar{R} \in C$$

$z = x + iy$ with $y \Rightarrow$
 ↓

If $z = x + iy$ and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$, then
show that $x^2 + y^2 = 1$ ✓

$$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1} = \frac{(x-1)+iy}{(x+1)+iy}$$

$$= \frac{(x-1)+iy}{(x+1)+iy} \times \frac{[(x+1)-iy]}{(x+1)-iy}$$

$$= \frac{(x-1)[(x+1)-iy] + iy[(x+1)-iy]}{(x+1)^2 - i^2 y^2}$$

$$= \frac{x^2 - 1 - iy(x-1) + iy(x+1) - i^2 y^2}{(x+1)^2 + y^2}$$

$$= \frac{x^2 - 1 - \cancel{ixy} + iy + \cancel{ixy} + iy + y^2}{(x+1)^2 + y^2}$$

$$= \frac{(x^2 + y^2 - 1) + 2iy}{(x+1)^2 + y^2}$$

$$\frac{z-1}{z+1} = \frac{x^2 + y^2 - 1}{(x+1)^2 + y^2} + i \frac{2y}{(x+1)^2 + y^2}$$

Because $\text{Arg}\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$

it shows that $\frac{z-1}{z+1}$ is a

pure imaginary number with

$$\text{Real part} = 0$$

$$\text{Arg}\left(\frac{z-1}{z+1}\right) = \tan^{-1}\left(\frac{\text{Im. part}}{\text{Real part}}\right) = \frac{\pi}{2}$$

Thus $\text{Real}\left(\frac{z-1}{z+1}\right) = 0$

$$\underline{x^2 + y^2 - 1 = 0}$$

$$\frac{x^2 + y^2 - 1}{(x+1)^2 + y^2} = 0$$

$$x^2 + y^2 - 1 = 0$$

$$\Rightarrow \boxed{x^2 + y^2 = 1}$$

Hence proved

Find the value of r , if
 $P_{r+6}^{56} : P_{r+3}^{54} = 30800 : 1$

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ \hline 1 & 4 & 3 & 2 \end{array} \checkmark$$

Sol

$${}_n P_r = \frac{n!}{(n-r)!}$$

$${}_n C_r = \frac{n!}{r!(n-r)!}$$

$$r! {}_n C_r = \frac{n!}{(n-r)!}$$

$${}_n C_r \times r! = {}_n P_r$$

Find the value of r , if
 $P_{r+6}^{56} : P_{r+3}^{54} = 30800 : 1$

$$56 - (r+6)$$

Sol

$$\frac{56!}{(56 - (r+6))!} : \frac{54!}{(54 - (r+3))!} = 30800 : 1$$

54
2

$$(56-2-6)!$$

$$(54-2-3)!$$

$$\frac{56!}{(50-r)!} : \frac{54!}{(\underline{51-r})!} = 30800:1$$

$$\frac{56 \cdot 55 \cdot 54!}{(50-r)!} : \frac{54!}{(51-r)(51-r-1)!} = 30800:1$$

$$(\because a:b \Rightarrow \frac{a}{b} \text{ or } a \div b)$$

$$\frac{56 \cdot 55 \cdot 54!}{(50-r)!} \div \frac{54!}{(51-r)(50-r)!} = \frac{30800}{1}$$

$$\frac{56 \cdot 55 \cdot \cancel{54!}}{(\cancel{50-r})!} \times \frac{(51-r)(\cancel{50-r})!}{\cancel{54!}} = 30800$$

$$56 \cdot 55 (51-r) = 30,800$$

$$51-r = \frac{30,800}{56 \cdot 55}$$

$$51 - r = 10$$

$$51 - 10 = r$$

$$\Rightarrow \boxed{r = 41} \quad \neq$$

Use Binomial Theorem to find the remainder when 5^{99} is divided by 13.

V. Imp $-(-x)$

$\Rightarrow$

$4 \rightarrow$ Quotient

$6 \sqrt{25} \rightarrow$ Divisor

24

$1 \rightarrow$ Remainder

$\frac{25}{6} = 4 \frac{1}{6}$

$25 = 6 \times 4 + 1$

Use Binomial Theorem to find the remainder when 5^{99} is divided by 13.

$(a+b)^n$

$\frac{98}{2} = 49$

$5^{99} = 5 \cdot 5^{98} = 5 \cdot 5^{2 \times 49} = 5 \cdot (5^2)^{49}$

$= 5 \cdot (25)^{49} = 5 \cdot (26 + (-1))^{49}$

$$= 5 \cdot (25) = 5 \cdot (\underline{\underline{26}} + (-1))$$

$$\frac{5^{99}}{13} = \frac{5 \cdot (26 + (-1))^{49}}{13}$$

$$\stackrel{\binom{n}{r} - \binom{n}{n-r}}{=} \frac{5}{13} \left[\binom{49}{0} 26^{49} + \dots + \binom{49}{48} 26^1 (-1) + \underline{\underline{\binom{49}{49} (-1)^{49}}} \right]$$

$$= \frac{5}{13} [13K - 1]$$

$$= \frac{1}{13} \left[\underline{\underline{5 \times 13K}} - 5 \right] \stackrel{\substack{Q \times Div + R \\ Div.}}{=}$$

$$= \frac{1}{13} \left[5 \times 13K - 13 + \underline{\underline{13 - 5}} \right]$$

$$= \frac{1}{13} \left[13(5K - 1) + 8 \right]$$

$$\frac{5^{99}}{13} = \frac{13(5K - 1) + 8}{13}$$

$$\boxed{\text{Remainder} = 8}$$

$$7^2 = \textcircled{49}$$

Example 14: Find the remainder when 7^{103} is divided by 25.

$$\frac{7^{103}}{25} = \frac{7 \cdot 7^{102}}{25} = \frac{7 \cdot (7^2)^{51}}{25}$$

$$\frac{7^{103}}{25} = \frac{7(49)^{51}}{25} = \frac{7(50 + (-1))^{51}}{25}$$

$$= \frac{7}{25} \left[\binom{51}{0} 50^{51} + \dots + \binom{51}{51} (-1)^{51} \right]$$

$$= \frac{7}{25} [25k - 1]$$

$$= \frac{1}{25} [7 \times 25k - 7]$$

$$= \frac{1}{25} [7 \times 25k - 25 + 25 - 7]$$

$$= \frac{1}{25} \left[7 \times 25K - 25 + \underline{\underline{25 - 7}} \right]$$

$$= \frac{1}{25} \left[\underline{\underline{25(7K-1)}} + \underset{\substack{\downarrow \\ \text{Re}}}{18} \right]$$

$$\Rightarrow \frac{7^{103}}{25} = \frac{25(7K-1) + 18}{25}$$

$$\boxed{\text{Remainder} = 18}$$

Find roots of the cubic polynomial
 $P(x) = 3x^3 - 5x^2 - 11x - 3$

$$1 \frac{2}{3} - 1 \frac{3}{2}$$

sol factor of constant term (P) are

$$P: \pm 1, \pm 3$$

Factors of leading coefficient (Q) are

$$Q: \pm 1, \pm 3$$

Thus, possible rational factors are

$$1, \pm 3$$

10ms, 10ms

$$\frac{p}{q} : \pm 1, \pm \frac{1}{3}, \pm 3,$$

By checking, $x = -1$ satisfies the given polynomial, hence it one root (zero) of $p(x)$

$$\boxed{x = -1}$$

By synthetic divs

$$\begin{array}{r|rrrr} -1 & 3 & -5 & -11 & -3 \\ & \downarrow & & & \\ & 3 & -8 & -3 & 0 \end{array}$$

$$Q(x) = 3x^2 - 8x - 3$$

$$P(x) = (x+1)(\underline{3x^2 - 8x - 3})$$

Now factorize $3x^2 - 8x - 3$

$$3x^2 - 8x - 3$$

$$\begin{array}{r} 4 \\ 6 \overline{) 24} \\ \underline{24} \\ 0 \end{array}$$

$$\begin{array}{l} x = -\frac{1}{3} \\ x = 3 \end{array}$$

$$\begin{array}{r} (x-3)(3x+1) \\ \underline{3x^2 + x - 9x - 3} \\ -3 \end{array}$$

$$\begin{aligned}
 &= 3x^2 + x - 9x - 3 \\
 &= x(3x+1) - 3(3x+1) \\
 &= (3x+1)(x-3)
 \end{aligned}$$

Thus Eqn (1) is

$$p(x) = (x+1)(x-3)(3x+1)$$

Thus, roots of $p(x)$ are

$$x+1=0 \Rightarrow \boxed{x=-1}$$

$$x-3=0 \Rightarrow \boxed{x=3}$$

$$3x+1=0 \Rightarrow \boxed{x=-\frac{1}{3}}$$

$$S.S = \left\{ -1, 3, -\frac{1}{3} \right\}$$

✓ If $A = \begin{bmatrix} x & 0 \\ y & 1 \end{bmatrix}$, then show that

$$A^n = \begin{bmatrix} x^n & 0 \\ y(x^n-1) & 1 \end{bmatrix}, n \in \mathbb{Z}^+$$

$p(n):$ $\begin{bmatrix} x^n & 0 \end{bmatrix}$

$$A^n = \begin{bmatrix} y(x^n-1) & 0 \\ (x-1) & 1 \end{bmatrix}, n \in \mathbb{Z}^+ \quad \text{given } P(n): A^n = \begin{bmatrix} x^n & 0 \\ \frac{y(x^n-1)}{x-1} & 1 \end{bmatrix}$$

SU1 Step-1 (Basis step)

Let $n=1$, $P(1)$ becomes

$$P(1): A^1 = A = \begin{bmatrix} x & 0 \\ \frac{y(x-1)}{x-1} & 1 \end{bmatrix} = \begin{bmatrix} x & 0 \\ y & 1 \end{bmatrix}$$

So, $P(n)$ is true for $n=1$

Step 2: (inductive step)

We assume that $P(k)$ is true
for $n=k$, thus $P(k+1)$
must also be true

$$\text{For } n=k, \quad P(k): A^k = \begin{bmatrix} x^k & 0 \\ \frac{y(x^k-1)}{x-1} & 1 \end{bmatrix}$$

for $n=k+1$, A^{k+1} must also be true

$$P(k+1): A^{k+1} = A^k \cdot A = \begin{bmatrix} x^k & 0 \\ \frac{y(x^k-1)}{x-1} & 1 \end{bmatrix} \begin{bmatrix} x & 0 \\ y & 1 \end{bmatrix}$$

$$= \begin{bmatrix} x^k \cdot x & 0 \\ xy \frac{(x^k-1)}{x-1} + y & 1 \end{bmatrix}$$

$$A^{k+1} = \begin{bmatrix} x^{k+1} & 0 \\ \frac{xy(x^k-1)}{x-1} + y & 1 \end{bmatrix} \rightarrow \textcircled{1}$$

Now, *simplify*

$$\frac{xy(x^k-1)}{x-1} + y$$

$$= \frac{xy(x^k-1) + y(x-1)}{x-1}$$

$$= \frac{y \cdot x^{k+1} - yx + yx - y}{x-1}$$

$$= \frac{4(x^{k+1} - 1)}{x - 1}$$

Thus Eqn (1) becomes

$$P(k+1) : A^{k+1} = \begin{bmatrix} x^{k+1} & 0 \\ \frac{4(x^{k+1} - 1)}{x - 1} & 1 \end{bmatrix}$$

Thus, the statement $P(k+1)$ is true for $n = k+1$ &

Hence $P(n)$ is true for all $n \in \mathbb{Z}^+$

Apply the principle of Mathematical Induction to prove that $7^{2n} + 7$ is divisible by 8 for all positive integral values of n .

$n \in \mathbb{Z}^+$

3

8 $\sqrt{24}$

24

Zero

Sol Given $P(n) : 7^{2n} + 7$

Step-1 put $n = 1$

2

< 1

Step 1

$$P(1) : 7^2 + 7 = 56$$

As 56 is clearly divisible
by 8. Hence statement
is for $n=1$ ✓

Step-2 Assume that $P(n)$ is
true for $n=k$, so, $P(k+1)$
must be true

$$\text{for } n=k, P(k) : 7^{2k} + 7$$

$$\text{Let } 7^{2k} + 7 = \underline{8Q} \rightarrow (1)$$

$$\text{Where } Q \text{ is any integer} \Rightarrow 7^{2k} = 8Q - 7 \rightarrow (2)$$

$$\text{for } n=k+1, P(k+1) : 7^{2(k+1)} + 7$$

$$7^{2k+2} + 7$$

$$\begin{aligned}
 P(k+1) &= 7^{2k+2} + 7 \\
 &= 7^{2k} \cdot 7^2 + 7 \\
 &= (8q-7)7^2 + 7 \quad (\text{from Eqn (2)}) \\
 &= 392q - 343 + 7
 \end{aligned}$$

$$\begin{aligned}
 P(k+1) &= 392q - 336 \\
 &= \underset{\uparrow}{8} (\underline{49q - 42})
 \end{aligned}$$

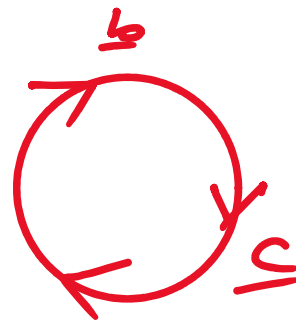
Thus, $P(k+1)$ is a multiple of 8, hence it is divisible

by 8. So, $P(k+1)$ is true.

Hence $P(n)$ is true for all positive integers.

If $\underline{a} = 2\underline{i} - \underline{j} + 3\underline{k}$, $\underline{b} = 3\underline{i} + 2\underline{j} + 4\underline{k}$ and $\underline{c} = \underline{i} + 3\underline{j} - 5\underline{k}$, then verify that

$$\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \boxed{\underline{c} \cdot \underline{a} \times \underline{b}}$$



$$\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \underline{c} \cdot \underline{a} \times \underline{b}$$

$|\underline{a} \cdot \underline{b} \times \underline{c}| = \text{Volume of parallelepiped}$

$\rightarrow$ scalar triple product

$$\underline{c} \cdot \underline{a} \times \underline{b} = \begin{vmatrix} 1 & 3 & -5 \\ 2 & -1 & 3 \\ 3 & 2 & 4 \end{vmatrix}$$

$$\begin{aligned} &= 1(-4-6) - 3(8-9) - 5(\underline{4+3}) \\ &= -10 + 3 - 35 \\ &= -42 \end{aligned}$$

Find the number of ways to select 3 balls from a collection of 4 orange, 5 red, and 6 green balls, such that:

Selection $\rightarrow$

from a collection of 4 orange, 5 red, and 6 green balls, such that:

- (a) All balls are of different colors. ✓
- (b) All balls are of the same color.
- (c) No ball is red.
- (d) Exactly one ball is green.

~~selection~~
Combinator

5 rrrrr
12

n
 $\binom{n}{r}$
3

$O_1 R_2 G_3$
 $G_1 O_2 R_3$

Sol
O 4
R 5
G 6
Total balls = 16

q)

No. of ways

for selecting ball of diff. colors

$$\binom{8}{1} = \binom{8}{7} = 8$$

$$= \binom{4}{1} \times \binom{5}{1} \times \binom{6}{1}$$

$$= 4 \times 5 \times 6 = 120 \text{ ways}$$

Find the number of ways to select 3 balls from a collection of 4 orange, 5 red, and 6 green balls, such that:

- (a) All balls are of different colors. ✓
- (b) All balls are of the same color. ✓
- (c) No ball is red.
- (d) Exactly one ball is green.

$O_1 O_2 O_3$ ✓
 $G_1 G_2 G_3$ ✓
 $R_1 R_2 R_3$ ✓

$$b) \text{ No. of ways} = \binom{4}{3} + \binom{5}{3} + \binom{6}{3}$$

$$= 4 + 10 + 20$$

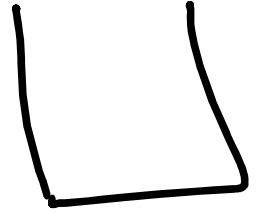
$$= 4 + 10 + 1 = 15$$

$$= 34 \checkmark$$

c)

Find the number of ways to select 3 balls from a collection of 4 orange, 5 red, and 6 green balls, such that:

- (a) All balls are of different colors.
- (b) All balls are of the same color.
- (c) No ball is red. $\checkmark$
- (d) Exactly one ball is green.



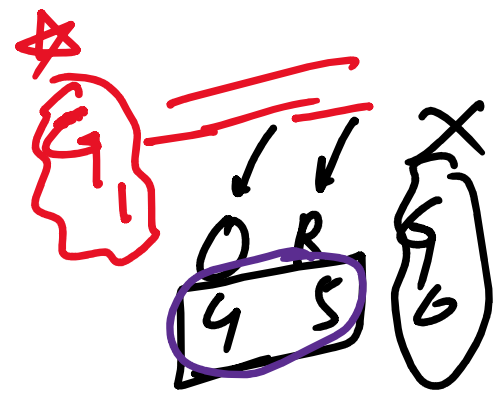
O R G
4 5 6

Total balls except red = 10

$$\text{No. of ways} = \binom{10}{3} = 120$$

Find the number of ways to select 3 balls from a collection of 4 orange, 5 red, and 6 green balls, such that:

- (a) All balls are of different colors.
- (b) All balls are of the same color.
- (c) No ball is red. $\checkmark$
- (d) Exactly one ball is green. $\checkmark$



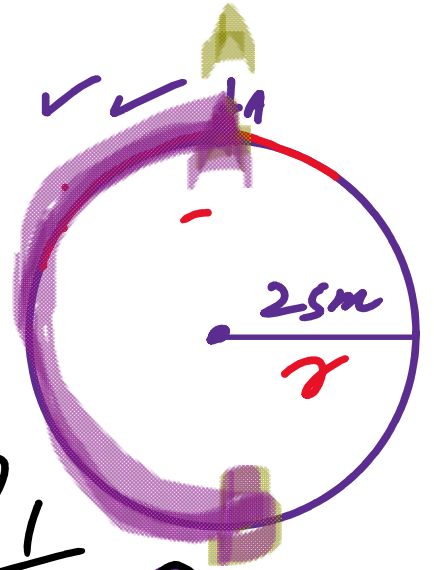
$$\text{d) No. of ways} = \binom{6}{1} \times \binom{9}{2}$$

$$= 6 \times 36$$

$$= 216 \text{ ways}$$

= 216 ways

A Ferris wheel with a radius of 25 meters completes one full revolution in 4 minutes. Calculate the frequency of the Ferris wheel's rotation, the speed of a passenger at the edge of the wheel, and the time it takes for the passenger to travel from the bottom to the top of the wheel.



a) Frequency = $\frac{1}{\text{Time period}} = \frac{1}{T}$

Given that Time period = $T = 4 \text{ minutes}$

$\Rightarrow T = 4 \times 60 \text{ sec} = 240 \text{ s}$

$\Rightarrow f = \frac{1}{T} = \frac{1}{240 \text{ s}} = 0.0042 \text{ Hz}$

b) Speed = $\frac{\text{distance covered}}{\text{Time taken}}$

$V = \frac{S}{T} = \frac{2\pi r}{240 \text{ s}} = \frac{2\pi \times 25 \text{ m}}{240 \text{ s}}$

$$V = 0.654 \text{ m/s}$$

c) Time taken = $\frac{T}{2} = 120 \text{ sec}$

Solve the following system of non-homogeneous linear equations by Gaussian Elimination Method:

$$2x - 3y + 5z = 2,$$

$$x + 4y - 2z = 1,$$

$$4x + 5y + z = 4$$

A_b
Echelon form

Sol $A_b = \begin{bmatrix} 2 & -3 & 5 & : & 2 \\ 1 & 4 & -2 & : & 1 \\ 4 & 5 & 1 & : & 4 \end{bmatrix}$

$R_1 \sim \begin{bmatrix} 1 & 4 & -2 & : & 1 \\ 2 & -3 & 5 & : & 2 \\ 4 & 5 & 1 & : & 4 \end{bmatrix}$ B_1
 $R_1 \leftrightarrow R_2$

$R_2 \sim \begin{bmatrix} 1 & 4 & -2 & : & 1 \\ 0 & -11 & 9 & : & 0 \\ 0 & -11 & 9 & : & 0 \end{bmatrix}$ B_2
 $R_2 + (-2)R_1 \rightarrow R_2'$
 $R_3 + (-4)R_1 \rightarrow R_3'$

$$\sim \begin{bmatrix} 0 & -11 & 9 & 0 \end{bmatrix} R_3 + (-4)R_1 \rightarrow R_3$$

$$\sim \begin{bmatrix} 1 & 4 & -2 & : & 1 \\ 0 & 1 & -\frac{9}{11} & : & 0 \\ 0 & 0 & -\frac{9}{11} & : & 0 \end{bmatrix} \begin{array}{l} \text{By} \\ (-\frac{1}{11})R_2 \rightarrow R_2' \\ (-\frac{1}{11})R_3 \rightarrow R_3' \end{array}$$

$$\sim \begin{bmatrix} 1 & 4 & -2 & : & 1 \\ 0 & 1 & -\frac{9}{11} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix} \begin{array}{l} \text{By} \\ R_3 + (-1)R_2 \rightarrow R_3' \end{array}$$

As Rank of $A = \text{Rank of } A_b = 2$

The system has consistent solution but as $\text{Rank} < \text{No. of unknowns}$

So, it has infinite many solutions

$$\text{So, } x + 4y - 2z = 1 \rightarrow (1)$$

$$y - \frac{9}{11}z = 0 \rightarrow (2)$$

$$0z = 0 \checkmark$$

Let $\boxed{z = t}$

put value of z in (2), we get

$$y - \frac{9}{11}(t) = 0$$

$$\boxed{y = \frac{9t}{11}}$$

Thus, Eqn (1) becomes,

$$x + 4\left(\frac{9t}{11}\right) - 2t = 1$$

$$11x + \underline{36t} - 22t = 11$$

$$11x + 14t = 11$$

$$11x = 11 - 14t$$

$$x = \frac{11 - 14t}{11}$$

$$\boxed{x = 1 - \frac{14}{11}t}$$

$$\boxed{x = 1 - \frac{4}{11}t}$$

$$S.S = \left\{ 1 - \frac{4}{11}t, \frac{9t}{11}, t \right\}$$

OR

$$\boxed{x = 1 - \frac{4}{11}t, y = \frac{9t}{11}, z = t}$$

$\underline{A_2}$

If x is very small such that its square and higher powers can be neglected, then show that

$$\frac{(8+3x)^{\frac{2}{3}}}{(2+3x)\sqrt{4-5x}} \approx 1 - \frac{5x}{8}$$

$\checkmark$ $-ve$ $\checkmark$ $\frac{1}{1+x} \approx 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$

$$\begin{aligned} L.H.S &= \frac{(8+3x)^{\frac{2}{3}}}{(2+3x)\sqrt{4-5x}} \\ &= \frac{(8+3x)^{\frac{2}{3}}}{(2+3x)(4-5x)^{\frac{1}{2}}} \end{aligned}$$

$$\begin{aligned}
& (2+3x)(4-5x)^2 \\
&= \left(8 \cdot \left(\frac{8+3x}{8} \right) \right)^{\frac{2}{3}} \\
&= \frac{\left(2 \cdot \left(\frac{2+3x}{2} \right) \right) \left(4 \cdot \left(\frac{4-5x}{4} \right) \right)^{\frac{1}{2}}}{\left(8 \left(1 + \frac{3x}{8} \right) \right)^{\frac{2}{3}}} \quad (x \cdot 8)^n \\
&= \frac{2 \left(1 + \frac{3x}{2} \right) \left(4 \left(1 - \frac{5x}{4} \right) \right)^{\frac{1}{2}}}{8^{\frac{2}{3}} \cdot \left(1 + \frac{3x}{8} \right)^{\frac{2}{3}}} \\
&= \frac{2 \left(1 + \frac{3x}{2} \right) 4^{\frac{1}{2}} \left(1 - \frac{5x}{4} \right)^{\frac{1}{2}}}{4 \cdot \left(1 + \frac{3x}{8} \times \frac{2}{3} \right)} \\
&= \frac{2 \cancel{2} \left(1 + \frac{3x}{2} \right) \left(1 - \frac{1}{2} \cdot \frac{5x}{4} \right)}{2 \cancel{2} \left(1 + \frac{3x}{2} \right) \left(1 - \frac{1}{2} \cdot \frac{5x}{4} \right)}
\end{aligned}$$

$$\cancel{2} \cancel{2} \left(1 + \frac{x}{2}\right) \left(1 - \frac{x}{4}\right)$$

$$= \frac{1 + \frac{x}{4}}{\left(1 + \frac{3x}{2}\right) \left(1 - \frac{5x}{8}\right)}$$

$$= \frac{1 + \frac{x}{4}}{1 - \frac{5x}{8} + \frac{3x}{2}}$$

$$= \frac{1 + \frac{x}{4}}{1 + \frac{2x}{2} - \frac{5x}{8}}$$

$$\frac{10x - 5x}{8}$$

$$= \frac{\left(1 + \frac{x}{4}\right)}{\left(1 + \frac{7x}{8}\right)}$$

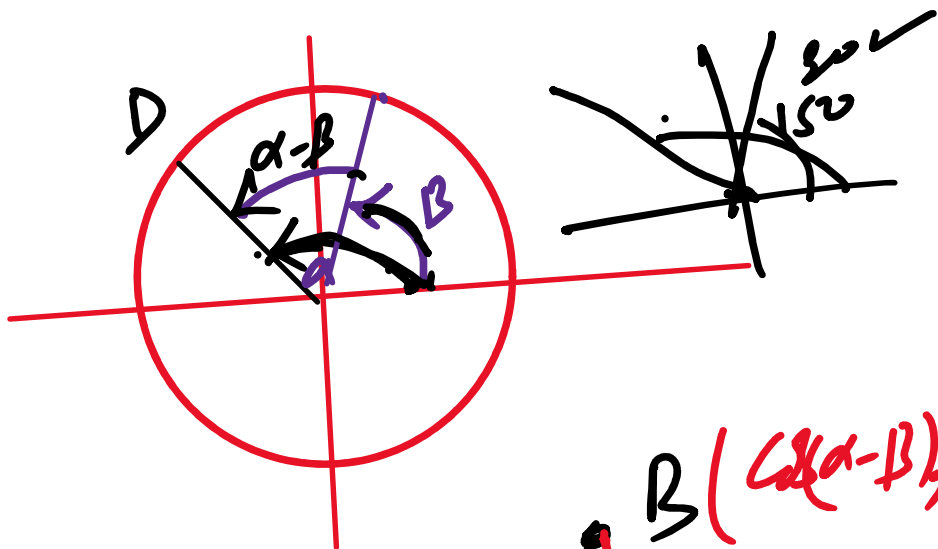
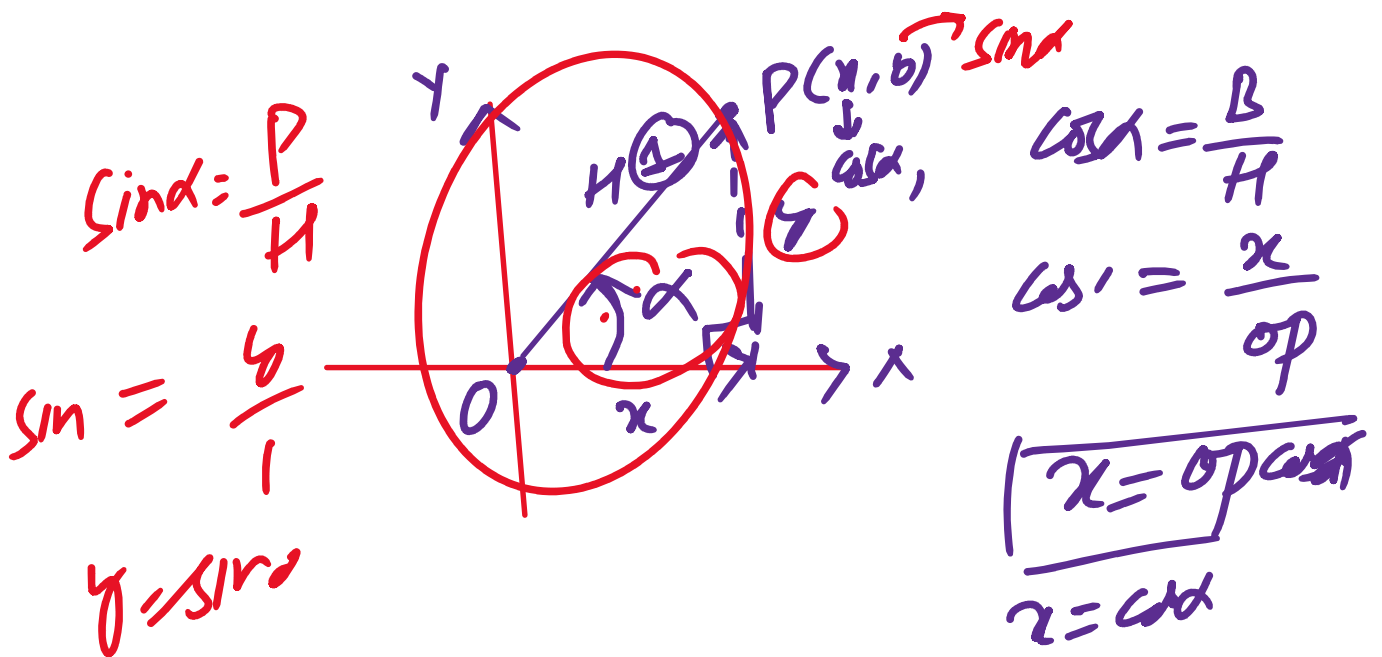
$$= \left(1 + \frac{x}{4}\right) \left(1 + \frac{7x}{8}\right)^{-1}$$

$$= \left(1 + \frac{x}{4}\right) \left(1 - \frac{7x}{8}\right)$$

$$= 1 - \frac{7}{8}x + \frac{x}{4}$$

$$= 1 - \frac{5}{8}x = \text{RHS}$$

Thus - prove



$$(a-b)^2$$

$$B(C(\alpha - \beta), \sin k - 1)$$

